

Sundry Asset Coverage and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Sundry Asset Coverage (SAC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on SAC achieves an annualized gross (net) Sharpe ratio of 0.32 (0.28), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (20) bps/month with a t-statistic of 3.23 (2.91), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market, original (Statman 1980), Book leverage (annual), Operating leverage, Equity Duration, Enterprise Multiple, Earnings-to-Price Ratio) is 18 bps/month with a t-statistic of 2.63.

1 Introduction

Financial markets efficiently incorporate information about firm fundamentals into asset prices, yet significant cross-sectional variation in expected returns persists even after controlling for standard risk factors. This variation reflects differences in firms' exposure to systematic risks arising from production decisions, investment frictions, and technological constraints that shape their cost structures and capital allocation choices. We examine how Sundry Asset Coverage (SAC), defined as the ratio of sundry assets to total assets, captures firms' production flexibility and operational efficiency in ways that predict future stock returns. Sundry assets—comprising miscellaneous operational assets not classified in major categories—represent firms' investments in specialized production inputs, intangible operational capabilities, and adaptive technologies that determine their ability to respond to productivity shocks. The economic significance of these assets stems from their role in determining marginal costs of production and the firm's capacity to adjust operations in response to changing economic conditions.

In production-based asset pricing models, cross-sectional variation in expected returns arises from heterogeneity in firms' technologies, adjustment costs, and exposure to aggregate productivity shocks [Cochrane and Piazzesi \(2005\)](#); [Zhang \(2005\)](#). Firms with higher sundry asset coverage maintain greater operational flexibility through diversified production inputs and specialized technologies that reduce adjustment costs when responding to demand or productivity shocks. These assets function as real options in the production process, enabling firms to modify output levels, switch between production technologies, or redeploy resources more efficiently than firms with rigid asset structures [Berk et al. \(1999\)](#). The marginal value of this flexibility increases during periods of high economic uncertainty when the option to adjust production becomes more valuable, leading to time-varying risk premia that compensate investors for bearing this systematic risk.

The production-based framework predicts that firms with higher SAC face lower operating leverage because sundry assets reduce the sensitivity of profits to revenue shocks through enhanced operational flexibility [Carlson et al. \(2004\)](#). When productivity shocks hit the economy, these firms can more readily adjust their cost structures by redeploying sundry assets across different production activities or scaling operations without incurring substantial adjustment costs. This operational flexibility manifests as lower systematic risk exposure during economic downturns, when the marginal utility of consumption is high and risk premia are elevated. Consequently, investors require lower expected returns to hold stocks of high-SAC firms, creating a negative relationship between sundry asset coverage and future returns in equilibrium [Zhang \(2005\)](#).

The link between SAC and expected returns also operates through the investment channel in production economies. Sundry assets represent past investments in operational capabilities that generate quasi-fixed costs, affecting the firm’s optimal investment policy and its response to investment opportunities [Liu et al. \(2009\)](#). Firms with substantial sundry assets have already incurred the sunk costs of building operational flexibility, reducing their need for costly external financing when positive NPV projects arise. This pre-existing flexibility lowers the shadow cost of capital and enables these firms to capture investment opportunities more efficiently. The resulting differences in investment-capital sensitivities across firms generate cross-sectional variation in risk exposures to aggregate investment shocks, with high-SAC firms exhibiting lower sensitivities and thus commanding lower risk premia in equilibrium [Lin and Zhang \(2013\)](#).

Our empirical analysis reveals that sundry asset coverage strongly predicts cross-sectional stock returns with economically and statistically significant results. A value-weighted long-short portfolio that buys stocks with high SAC and sells stocks with low SAC generates an average monthly return of 19 basis points with a t-

statistic of 2.49, corresponding to an annualized Sharpe ratio of 0.32. The strategy’s alpha remains robust across all standard factor models, with the six-factor alpha reaching 22 basis points per month (t-statistic = 3.23), confirming that SAC captures a distinct source of systematic risk not explained by existing factors. The strategy loads significantly negative on the value factor (HML) with a coefficient of -0.38 (t-statistic = -12.00), consistent with high-SAC firms exhibiting growth-like characteristics through their investments in operational flexibility.

The predictive power of SAC persists after accounting for transaction costs, with the net Sharpe ratio remaining at 0.28 and the net six-factor alpha equal to 20 basis points per month (t-statistic = 2.91). Among large-cap stocks exceeding the 80th NYSE size percentile, the SAC strategy delivers an average return of 15 basis points per month (t-statistic = 1.82), demonstrating that the anomaly is not confined to small, illiquid stocks where limits to arbitrage might prevent efficient pricing. The strategy’s performance ranks in the top third of the factor zoo, with its gross Sharpe ratio exceeding 67% of documented anomalies and its net Sharpe ratio surpassing 89% of anomalies after transaction costs.

Controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the SAC strategy maintains an alpha of 18 basis points per month with a t-statistic of 2.63. The Fama-MacBeth cross-sectional regressions confirm that SAC retains significant predictive power for returns even after controlling for related characteristics, with a coefficient of 0.78 (t-statistic = 2.62) in the full specification. These results establish that sundry asset coverage captures a unique dimension of firms’ production characteristics that generates systematic variation in expected returns beyond what existing factors and anomalies explain.

This paper contributes to the production-based asset pricing literature by identifying sundry asset coverage as a novel predictor of cross-sectional returns that directly measures firms’ operational flexibility and production efficiency. While [Fama](#)

and French (2015) document that investment and profitability factors help explain the cross-section of returns, and Hou et al. (2015) develop a q-factor model based on investment theory, our SAC measure provides a more direct proxy for the production flexibility that theory predicts should drive risk premia. Unlike broad investment measures that aggregate all capital expenditures, SAC specifically isolates the component of assets that provides operational optionality and reduces adjustment costs in production-based models. Our findings extend Berk et al. (1999) by empirically validating that operational flexibility embedded in sundry assets generates systematic risk variation consistent with real options theory in production economies.

Methodologically, we advance the literature by demonstrating that accounting classifications of sundry assets contain valuable information about firms’ production technologies and risk exposures not captured by standard financial ratios. Previous work such as Cooper et al. (2008) and Liu et al. (2009) emphasizes the importance of asset composition for understanding firms’ systematic risks, but focuses primarily on broad categories like fixed assets or intangibles. Our granular focus on sundry assets reveals that even seemingly minor balance sheet items can have first-order effects on expected returns when they capture fundamental aspects of firms’ production functions. This finding suggests that researchers should examine detailed accounting classifications more carefully to uncover production characteristics that drive cross-sectional return variation.

Our results have broader implications for understanding how production decisions and operational strategies affect firms’ costs of capital in equilibrium. The strong negative relationship between SAC and expected returns indicates that investors rationally price the value of operational flexibility, requiring lower returns from firms that maintain buffer assets to manage production uncertainty. This finding informs corporate finance decisions about optimal asset composition and suggests that managers face a trade-off between operational flexibility and cost of capital when choosing

their mix of specialized versus general-purpose assets. For asset pricing, our evidence supports production-based models' prediction that cross-sectional return variation reflects heterogeneity in firms' technologies and adjustment costs rather than purely financial frictions or behavioral biases.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of balance sheet items across firms. Sundry Asset Coverage signal combines two balance sheet components. Current assets - other sundry (ACOX) represents miscellaneous current assets that do not fall into standard categories such as cash, receivables, or inventory. These sundry assets typically include prepaid expenses, deferred charges, and other short-term assets that firms expect to convert to cash or consume within one operating cycle. Equity liquidation value (CEQL) measures the book value of common equity adjusted for the liquidation value of preferred stock, representing the residual claim of common shareholders on the firm's assets after all liabilities and preferred stock claims are satisfied. We construct the Sundry Asset Coverage signal by dividing current assets - other sundry (ACOX) by equity liquidation value (CEQL). This ratio captures the proportion of shareholders' residual claims that could theoretically be covered by miscellaneous current assets, providing insight into the composition and quality of a firm's asset base relative to its equity value. We calculate this measure annually using fiscal year-end values from firms' balance sheets. To ensure data quality, we require non-missing values for both ACOX and CEQL, and we exclude observations where CEQL equals

zero to avoid undefined ratios. The resulting signal provides a standardized measure that facilitates cross-sectional comparisons of firms' sundry asset positions relative to their equity base.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SAC signal. Panel A plots the time-series of the mean, median, and interquartile range for SAC. On average, the cross-sectional mean (median) SAC is 0.07 (0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input SAC data. The signal's interquartile range spans -0.00 to 0.10. Panel B of Figure 1 plots the time-series of the coverage of the SAC signal for the CRSP universe. On average, the SAC signal is available for 6.30% of CRSP names, which on average make up 7.31% of total market capitalization.

4 Does SAC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SAC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SAC portfolio and sells the low SAC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short SAC strategy earns an average return of 0.19% per month with a t-statistic of 2.49. The annualized Sharpe ratio of the strategy is 0.32. The alphas range from 0.20% to 0.31% per month and

have t-statistics exceeding 2.51 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.38, with a t-statistic of -12.00 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 559 stocks and an average market capitalization of at least \$1,537 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 3.11. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas

measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -1-19bps/month. The lowest return, (-1 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.11. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SAC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the SAC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SAC, as well as average returns and alphas for long/short trading SAC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the SAC strategy achieves an average return of 15 bps/month with a t-statistic of 1.82. Among these large cap stocks, the alphas for the SAC strategy relative to the five most common factor models range from 17 to 29 bps/month with t-statistics between 2.03 and 3.86.

5 How does SAC perform relative to the zoo?

Figure 2 puts the performance of SAC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212

documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the SAC strategy falls in the distribution. The SAC strategy’s gross (net) Sharpe ratio of 0.32 (0.28) is greater than 67% (89%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the SAC strategy (red line).² Ignoring trading costs, a \$1 invested in the SAC strategy would have yielded \$2.59 which ranks the SAC strategy in the top 11% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the SAC strategy would have yielded \$1.99 which ranks the SAC strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the SAC relative to those. Panel A shows that the SAC strategy gross alphas fall between the 42 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SAC strategy has a positive net generalized alpha for five out of the five factor models. In these cases SAC ranks between the 64 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does SAC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of SAC with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SAC or at least to weaken the power SAC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of SAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SAC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SAC}SAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SAC. Stocks are finally grouped into five SAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

SAC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SAC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SAC signal in these Fama-MacBeth regressions exceed 1.91, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on SAC is 2.62.

Similarly, Table 5 reports results from spanning tests that regress returns to the SAC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SAC strategy earns alphas that range from 20-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SAC trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.63.

7 Does SAC add relative to the whole zoo?

Finally, we can ask how much adding SAC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the SAC signal.⁴ We consider one different methods for combining signals.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SAC is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes SAC grows to \$2537.65.

8 Conclusion

This study provides compelling evidence that Sundry Asset Coverage (SAC) represents a robust and economically significant predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that SAC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.32 gross (0.28 net), confirming that the signal’s predictive power survives realistic implementation costs. Most notably, the strategy delivers statistically significant alpha of 22 basis points per month (t-statistic = 3.23) relative to the Fama-French five-factor model augmented with momentum, indicating that SAC captures unique information about expected returns not subsumed by conventional risk factors. The persistence of significant alpha (18 bps/month, t-statistic = 2.63) even after controlling for six closely related strategies—including traditional value measures, leverage metrics, and earnings-based ratios—underscores SAC’s incremental contribution to our understanding of the cross-section of returns. From a practical perspective, these findings suggest that SAC could serve as a valuable addition to quantitative investment strategies, particularly for investors seeking

exposure to factors beyond traditional value and quality metrics. The signal's economic magnitude and statistical robustness, combined with its distinctiveness from existing factors, make it a promising candidate for portfolio construction and risk management applications. However, several limitations warrant consideration. First, while our analysis accounts for transaction costs through net return calculations, real-world implementation may face additional frictions such as market impact and capacity constraints that could erode profitability. Second, the stability of SAC's predictive power across different market regimes and economic cycles requires further investigation. Third, the economic mechanism underlying SAC's return predictability remains an open question that deserves theoretical exploration. Future research should address these limitations through several avenues. Examining the time-varying nature of SAC's premium and its relationship to macroeconomic variables could provide insights into the risk-based versus mispricing interpretations of the anomaly. International evidence would help establish whether SAC's predictive power extends beyond U.S. markets, while analysis of the signal's performance in different size segments could reveal important heterogeneity. Additionally, investigating the fundamental economic drivers of SAC through detailed accounting analysis and firm-level case studies could enhance our understanding of why this measure successfully predicts returns. Finally, exploring potential improvements to the SAC signal through machine learning techniques or combination with other predictors may uncover additional alpha opportunities while maintaining the signal's robust statistical properties.

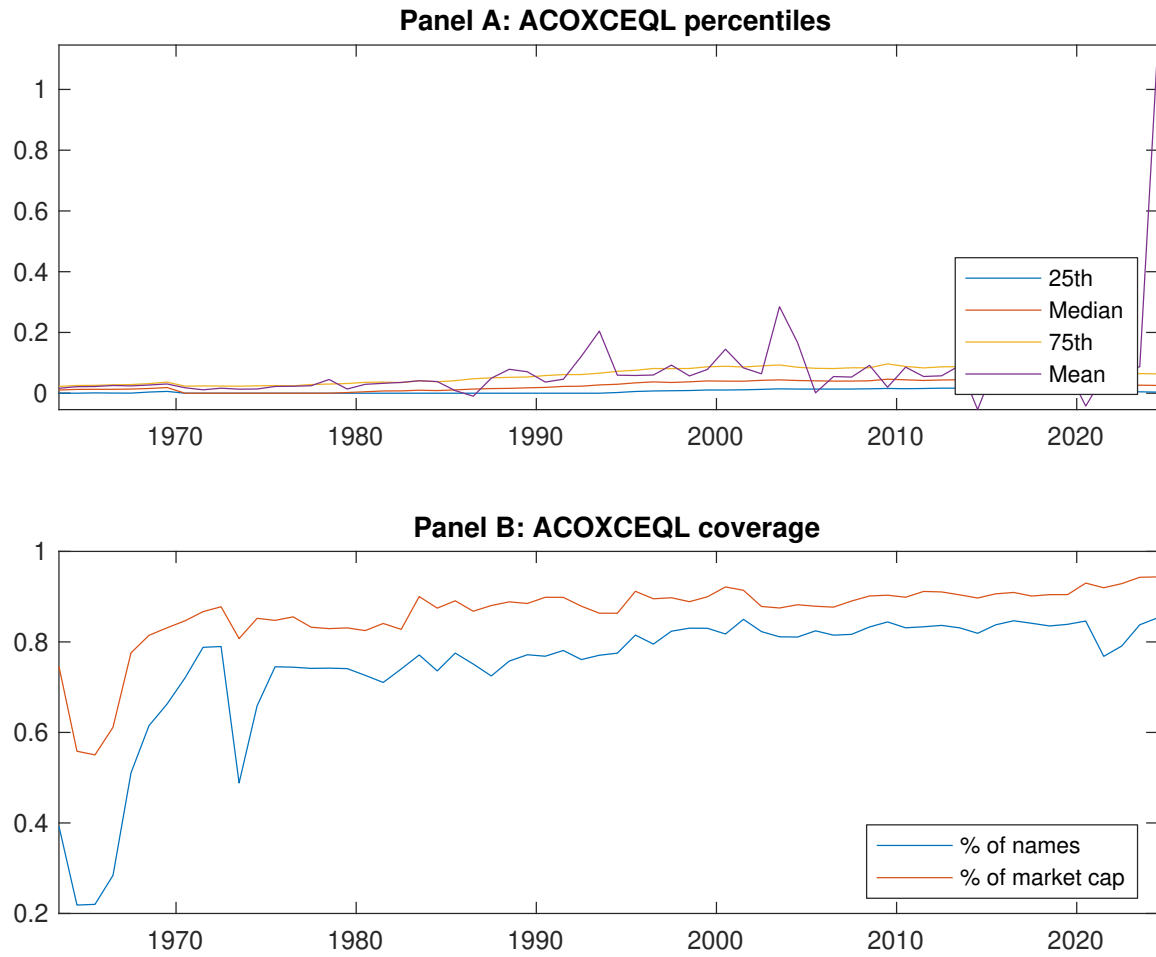


Figure 1: Times series of SAC percentiles and coverage.
This figure plots descriptive statistics for SAC. Panel A shows cross-sectional percentiles of SAC over the sample. Panel B plots the monthly coverage of SAC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SAC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on SAC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [3.14]	0.51 [2.87]	0.63 [3.72]	0.62 [3.69]	0.75 [4.29]	0.19 [2.49]
α_{CAPM}	-0.05 [-0.90]	-0.10 [-1.96]	0.05 [1.06]	0.04 [0.97]	0.15 [3.23]	0.20 [2.51]
α_{FF3}	-0.13 [-2.86]	-0.07 [-1.47]	0.07 [1.59]	0.09 [2.27]	0.19 [4.15]	0.31 [4.64]
α_{FF4}	-0.11 [-2.50]	-0.04 [-0.76]	0.04 [0.93]	0.08 [1.92]	0.17 [3.71]	0.28 [4.12]
α_{FF5}	-0.15 [-3.32]	-0.02 [-0.37]	0.05 [1.08]	0.08 [1.87]	0.09 [2.14]	0.24 [3.57]
α_{FF6}	-0.14 [-2.98]	0.01 [0.15]	0.03 [0.58]	0.07 [1.59]	0.08 [1.96]	0.22 [3.23]
Panel B: Fama and French (2018) 6-factor model loadings for SAC-sorted portfolios						
β_{MKT}	1.08 [96.74]	1.01 [83.14]	0.98 [92.87]	0.98 [96.65]	1.02 [98.95]	-0.06 [-3.45]
β_{SMB}	-0.07 [-4.44]	-0.03 [-1.76]	0.05 [3.37]	-0.00 [-0.16]	0.09 [6.18]	0.16 [6.85]
β_{HML}	0.22 [10.17]	-0.06 [-2.39]	-0.07 [-3.56]	-0.12 [-6.07]	-0.16 [-8.24]	-0.38 [-12.00]
β_{RMW}	0.05 [2.23]	-0.12 [-5.16]	0.04 [1.88]	0.06 [2.90]	0.22 [11.10]	0.18 [5.42]
β_{CMA}	0.04 [1.42]	-0.05 [-1.32]	0.02 [0.71]	-0.04 [-1.25]	0.10 [3.34]	0.05 [1.13]
β_{UMD}	-0.02 [-1.77]	-0.04 [-3.11]	0.03 [2.97]	0.02 [1.52]	0.01 [0.91]	0.03 [1.76]
Panel C: Average number of firms (n) and market capitalization (me)						
n	697	813	704	601	559	
me (\$10 ⁶)	1537	1623	1908	2200	2207	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the SAC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.19 [2.49]	0.20 [2.51]	0.31 [4.64]	0.28 [4.12]	0.24 [3.57]	0.22 [3.23]
Quintile	NYSE	EW	0.17 [3.11]	0.14 [2.50]	0.12 [2.28]	0.12 [2.21]	0.05 [1.03]	0.06 [1.08]
Quintile	Name	VW	0.22 [2.75]	0.22 [2.73]	0.34 [4.89]	0.30 [4.29]	0.26 [3.79]	0.24 [3.39]
Quintile	Cap	VW	0.19 [2.48]	0.20 [2.60]	0.31 [4.56]	0.28 [4.08]	0.24 [3.57]	0.22 [3.26]
Decile	NYSE	VW	0.18 [2.00]	0.18 [1.98]	0.27 [3.19]	0.25 [2.89]	0.25 [2.83]	0.23 [2.59]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.17 [2.17]	0.17 [2.21]	0.28 [4.08]	0.26 [3.81]	0.21 [3.09]	0.20 [2.91]
Quintile	NYSE	EW	-0.01 [-0.11]					
Quintile	Name	VW	0.19 [2.44]	0.20 [2.45]	0.30 [4.35]	0.28 [4.04]	0.23 [3.34]	0.21 [3.12]
Quintile	Cap	VW	0.16 [2.14]	0.17 [2.17]	0.26 [3.90]	0.25 [3.64]	0.20 [2.97]	0.19 [2.81]
Decile	NYSE	VW	0.15 [1.62]	0.15 [1.58]	0.23 [2.65]	0.22 [2.49]	0.19 [2.20]	0.18 [2.09]

Table 3: Conditional sort on size and SAC

This table presents results for conditional double sorts on size and SAC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SAC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SAC and short stocks with low SAC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	SAC Quintiles					SAC Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.61 [2.42]	0.54 [2.11]	0.80 [3.14]	0.88 [3.56]	0.84 [3.03]	0.23 [2.10]	0.19 [1.70]	0.14 [1.24]	0.13 [1.15]	0.07 [0.63]	0.07 [0.62]
	(2)	0.76 [3.26]	0.65 [2.74]	0.76 [3.20]	0.81 [3.45]	0.93 [3.82]	0.17 [2.08]	0.14 [1.73]	0.13 [1.58]	0.16 [1.92]	0.09 [1.02]	0.12 [1.36]
	(3)	0.72 [3.42]	0.65 [3.07]	0.77 [3.54]	0.82 [3.71]	0.93 [4.11]	0.21 [2.52]	0.16 [1.94]	0.16 [1.93]	0.15 [1.81]	0.13 [1.59]	0.13 [1.53]
	(4)	0.65 [3.19]	0.57 [2.93]	0.73 [3.61]	0.90 [4.47]	0.77 [3.74]	0.12 [1.50]	0.11 [1.35]	0.17 [2.06]	0.15 [1.80]	0.11 [1.29]	0.10 [1.16]
	(5)	0.56 [3.17]	0.55 [3.19]	0.55 [3.42]	0.54 [3.24]	0.71 [4.23]	0.15 [1.82]	0.17 [2.03]	0.29 [3.86]	0.26 [3.37]	0.22 [2.81]	0.20 [2.50]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	SAC Quintiles					SAC Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	378	386	388	387	380	28	31	33	32	29	
	(2)	104	104	104	104	103	53	54	54	54	54	
	(3)	72	72	73	73	72	92	90	92	92	94	
	(4)	60	60	60	60	60	199	192	196	197	204	
(5)	55	55	55	55	55	1347	1402	1470	1784	1604		

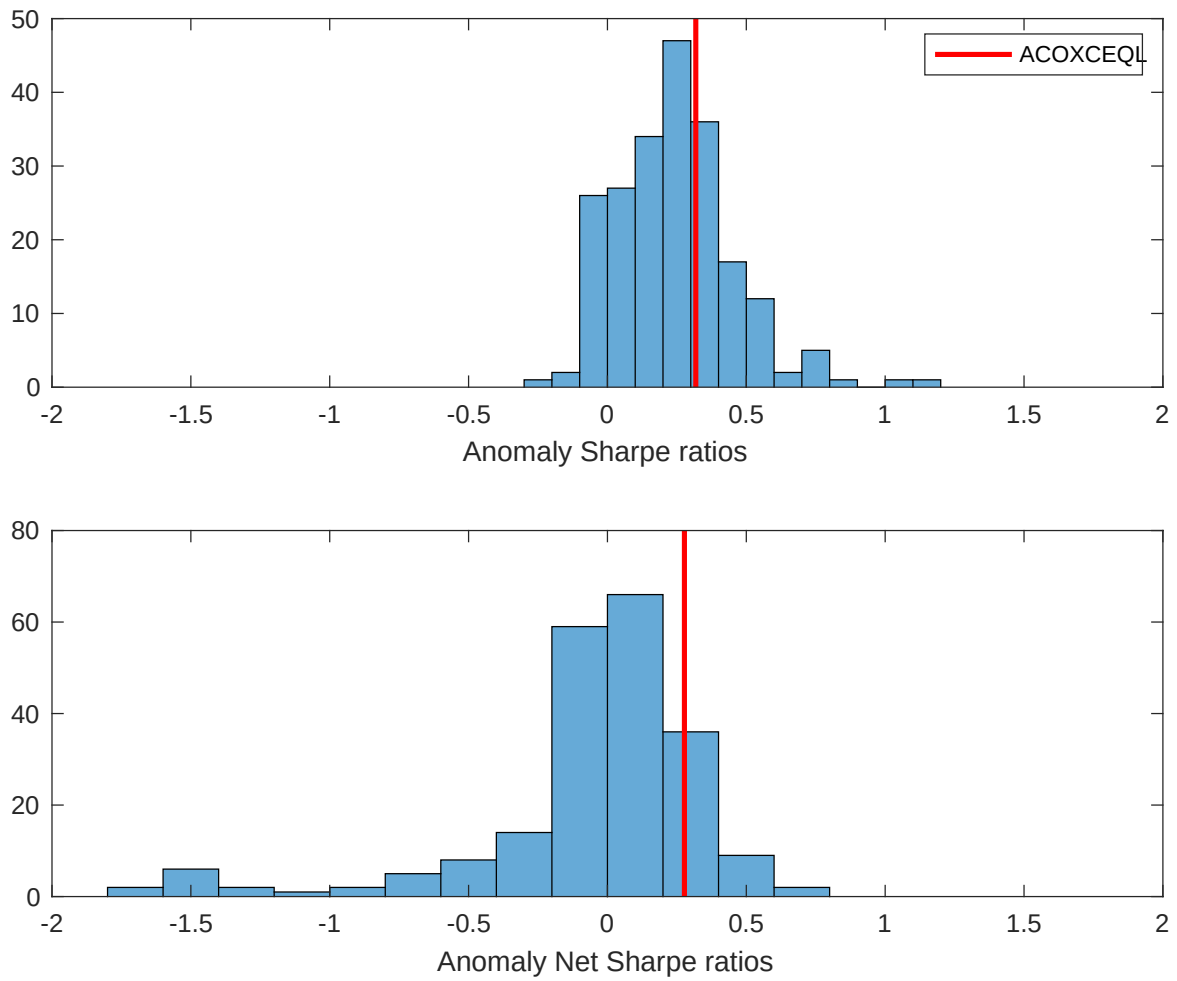


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SAC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

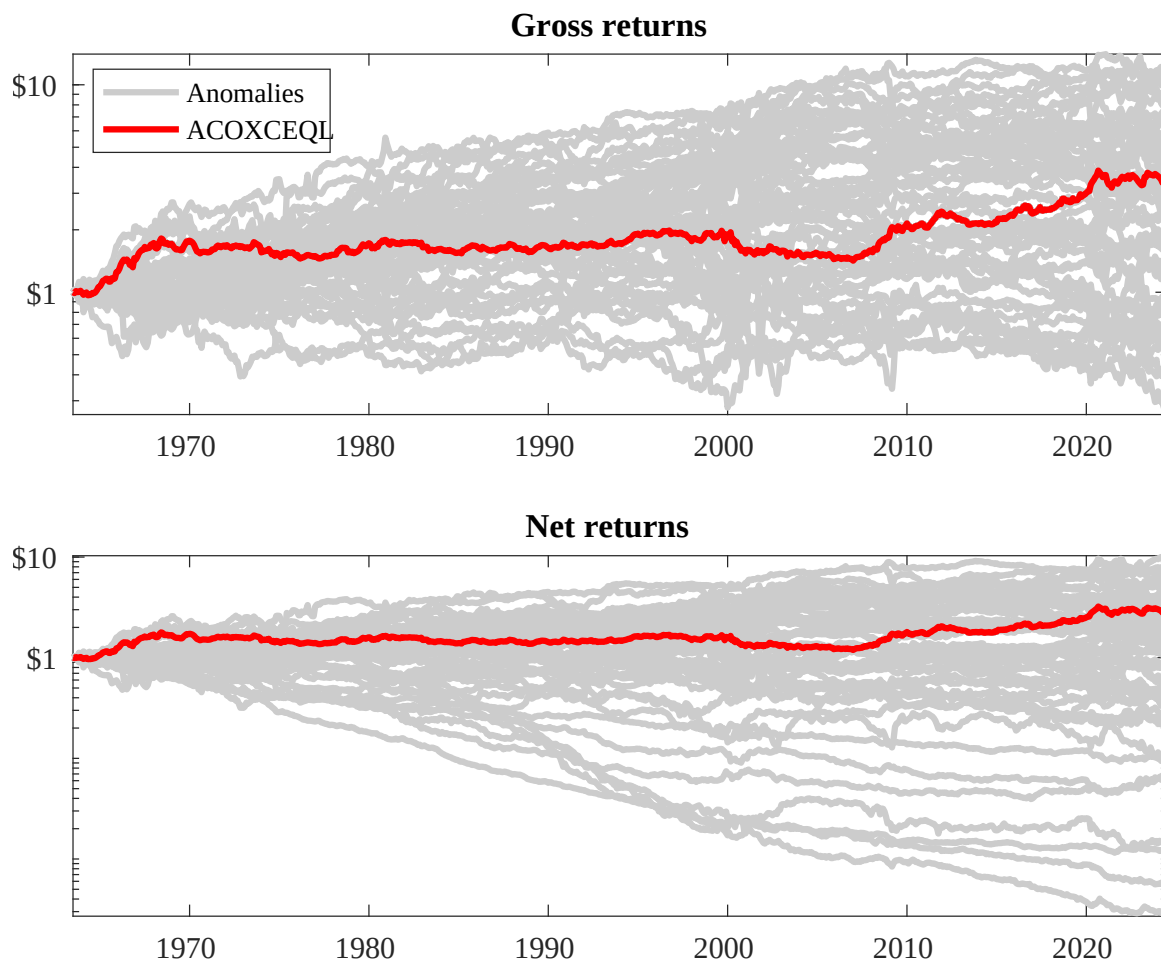
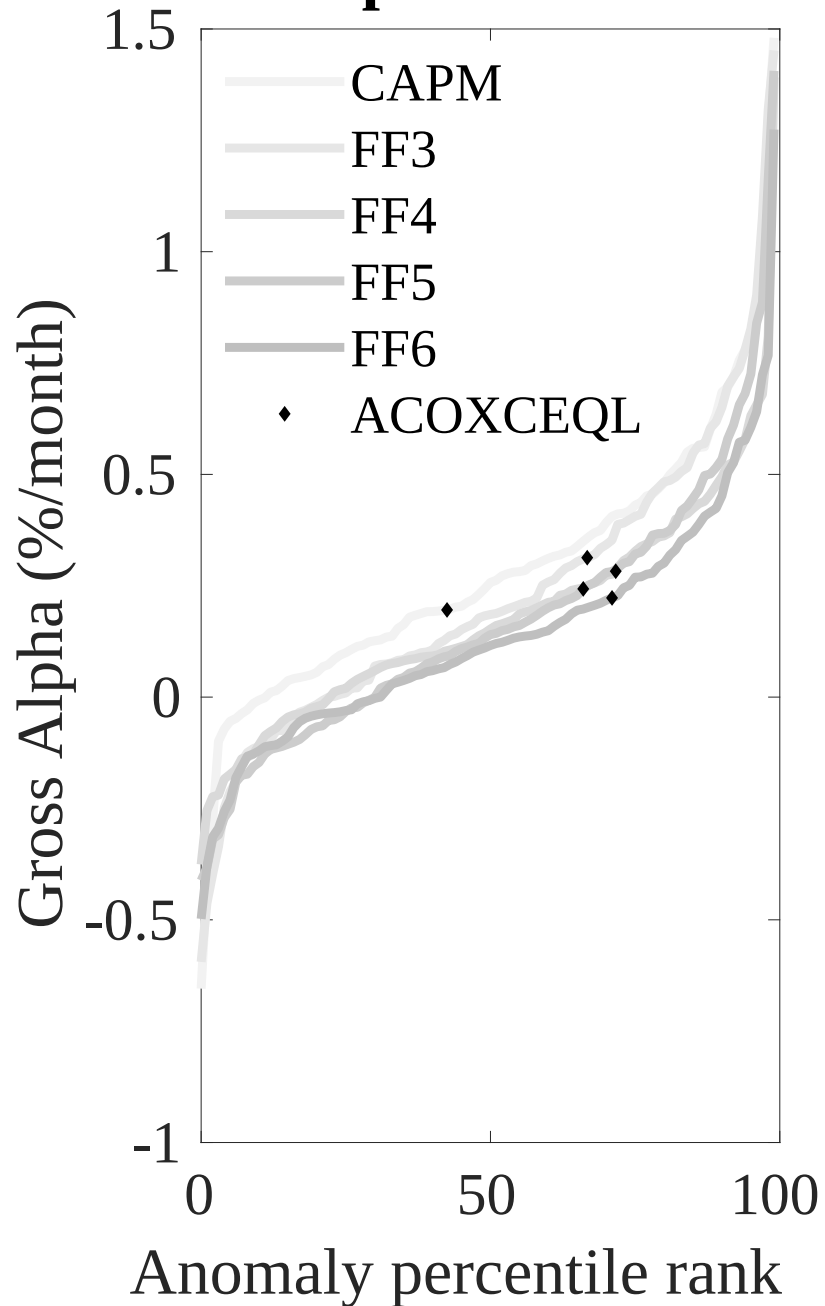


Figure 3: Dollar invested.

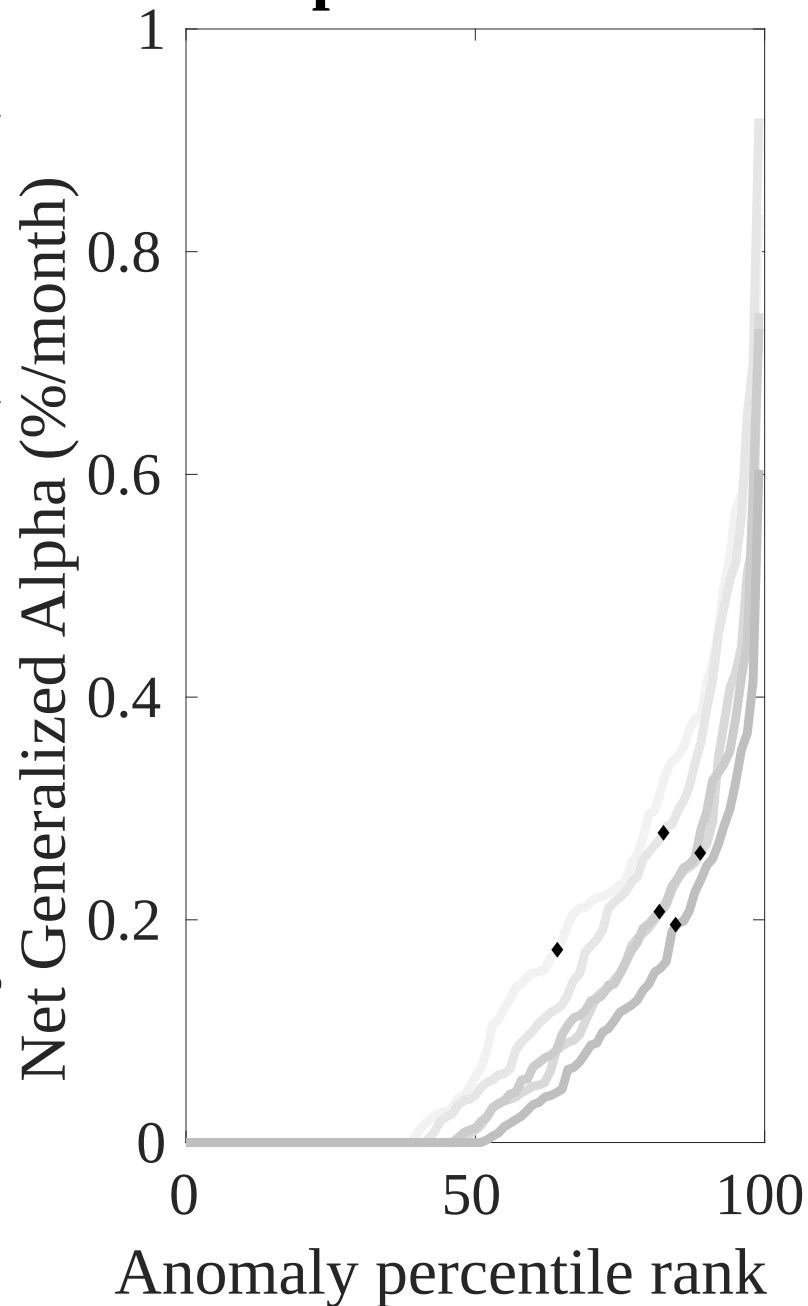
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SAC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



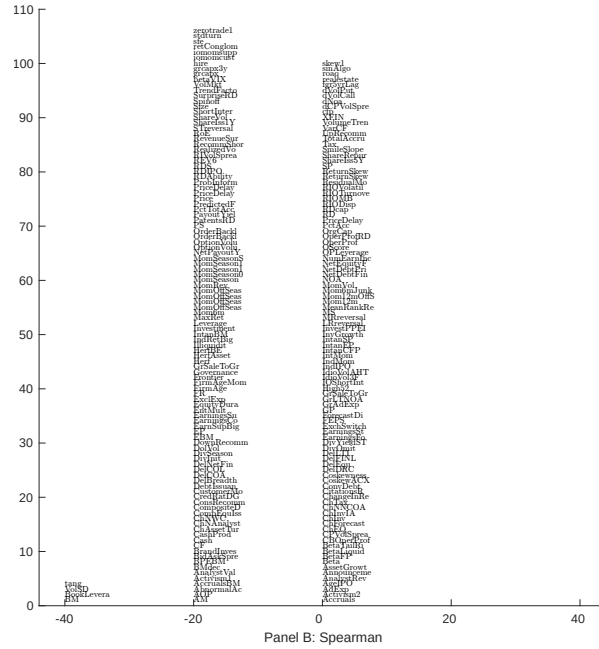
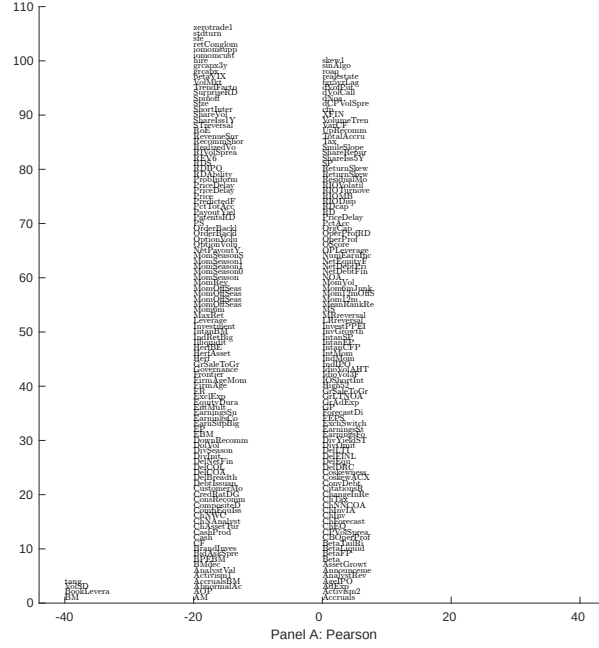


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with SAC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

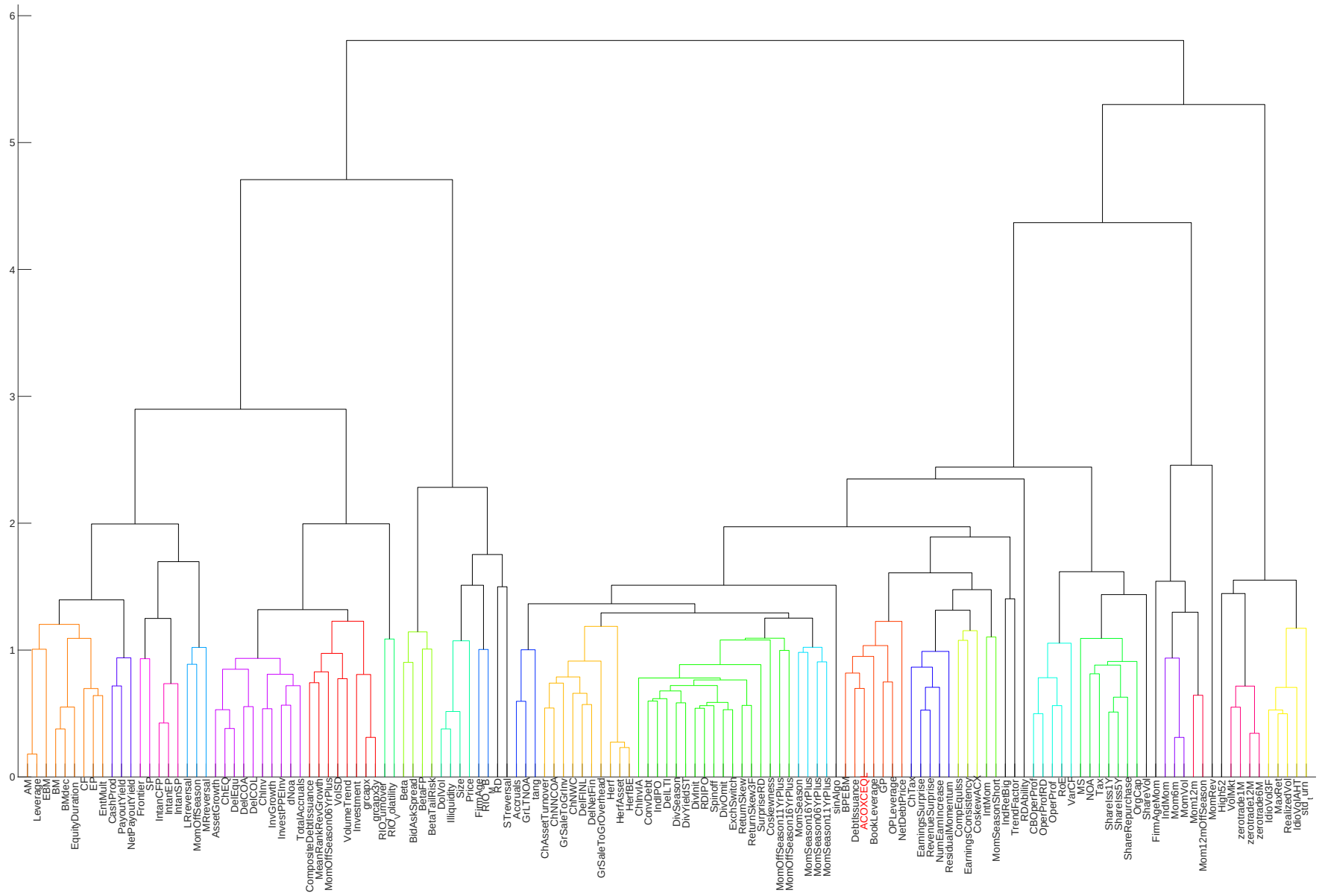


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

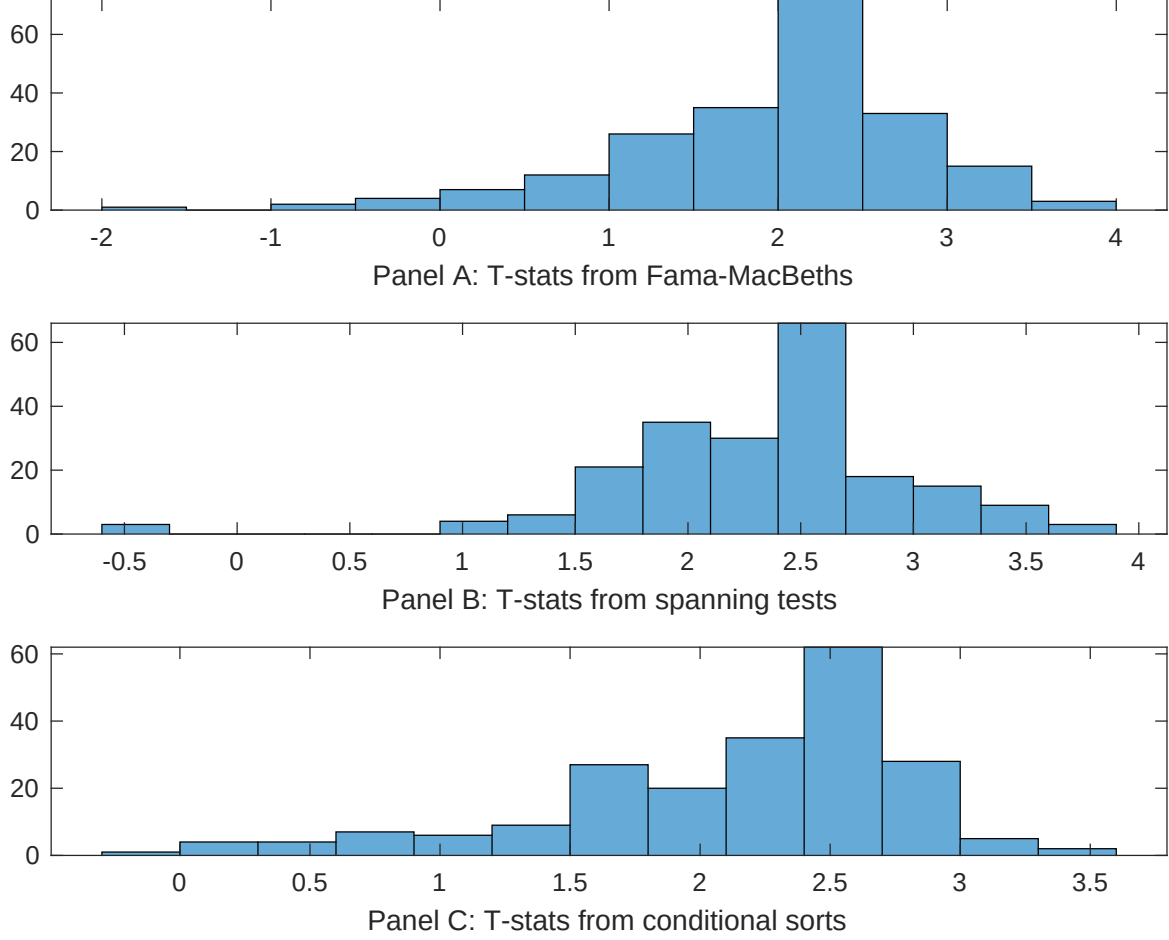


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of SAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SAC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SAC}SAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SAC. Stocks are finally grouped into five SAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SAC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SAC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SAC}SAC_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Enterprise Multiple, Earnings-to-Price Ratio. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.32]	0.12 [5.20]	0.10 [4.47]	0.20 [9.06]	0.13 [6.31]	0.11 [4.94]	0.21 [8.42]
SAC	0.11 [3.22]	0.92 [3.23]	0.60 [1.91]	0.10 [3.29]	0.78 [2.45]	0.79 [2.60]	0.78 [2.62]
Anomaly 1	0.38 [7.71]						0.90 [2.01]
Anomaly 2		0.16 [1.55]					-0.17 [-1.19]
Anomaly 3			0.11 [3.12]				0.87 [2.66]
Anomaly 4				0.52 [5.34]			0.58 [5.44]
Anomaly 5					0.81 [3.47]		1.00 [3.12]
Anomaly 6						0.17 [2.57]	-0.45 [-0.65]
# months	727	732	732	720	727	727	715
$\bar{R}^2(\%)$	1	0	0	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SAC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SAC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Book to market, original (Stattman 1980), Book leverage (annual), Operating leverage, Equity Duration, Enterprise Multiple, Earnings-to-Price Ratio. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.88]	0.20 [2.94]	0.20 [2.92]	0.22 [3.16]	0.25 [3.60]	0.22 [3.17]	0.18 [2.63]
Anomaly 1	-21.13 [-5.34]						-20.81 [-4.60]
Anomaly 2		6.86 [2.67]					4.38 [1.60]
Anomaly 3			18.48 [6.84]				17.31 [6.14]
Anomaly 4				-6.96 [-1.95]			-0.37 [-0.08]
Anomaly 5					-8.56 [-3.03]		-7.73 [-2.30]
Anomaly 6						-0.50 [-0.16]	13.82 [3.56]
mkt	-4.82 [-2.95]	-4.72 [-2.80]	-5.41 [-3.37]	-5.51 [-3.33]	-5.87 [-3.55]	-5.55 [-3.33]	-3.84 [-2.35]
smb	24.52 [8.84]	16.18 [6.77]	8.30 [3.17]	18.60 [7.15]	18.13 [7.45]	16.82 [6.78]	14.97 [4.97]
hml	-15.96 [-3.12]	-33.03 [-9.08]	-27.94 [-8.17]	-29.89 [-5.84]	-31.44 [-8.39]	-36.94 [-8.53]	-11.49 [-2.06]
rmw	14.03 [4.34]	19.35 [5.84]	7.10 [2.05]	17.09 [5.30]	18.12 [5.61]	17.33 [5.29]	4.50 [1.19]
cma	6.81 [1.47]	4.99 [1.07]	1.42 [0.31]	3.60 [0.76]	5.53 [1.18]	5.00 [1.06]	5.18 [1.12]
umd	4.08 [2.51]	2.77 [1.69]	2.70 [1.69]	2.99 [1.82]	0.08 [0.04]	3.04 [1.75]	3.20 [1.63]
# months	728	732	732	732	728	728	728
$\bar{R}^2(\%)$	31	30	33	29	30	29	36

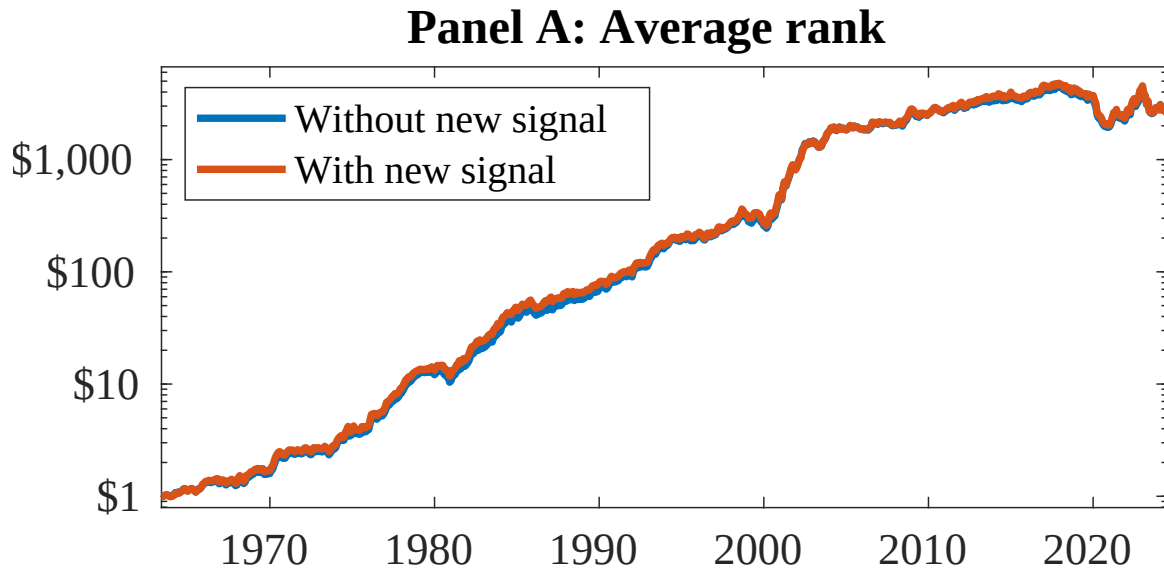


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as SAC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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